B. Project Overview

1. Brief Description of the Personal Data

This project tracks and analyzes performance data from a structured resistance training program between March 13th and April 5th, 2026. The dataset consists of 11 distinct training sessions recorded in an Excel environment, focusing on three foundational exercises: Hack Squat, Machine Chest Press, and Cable Row. The following metrics are tracked for each session:

- Weights (kg): The specific load used for the exercise.
- Repetitions: The count of completed repetitions per exercise.
- Total Work (Volume): Calculated as (Weights times Reps), representing the overall mechanical output of the session or the exercise.

2. Reasons for Using This Dataset

These three exercises were selected because they work the body's three largest muscle groups: legs, chest, and back. Monitoring these movements allows for a clear assessment of overall strength and endurance. By focusing on a few important exercises, the data stays consistent and makes it easier to track how performance improves over time.

3. Business-Style Framing

In this analysis, the workout routine is viewed as a production process. Performance is measured through three key categories:

- Weights (kg): Represents the difficulty level of each task.
- Repetitions: Shows the amount of work done for each task.
- Total Work: Serves as the main success goal and the final result of the training session.

The objective is to track how performance grows over time and to determine if lifting heavier weights leads to a higher total output

4. Stakeholder and Research Questions

Stakeholder: The Individual / Athlete

Research Questions:

- Progression: What is the percentage increase in "Total Work" for each exercise from the start to the end of the 11-session period?
- Performance Balance: Which of the three exercises shows the most consistent growth in weights versus repetitions?
- Correlation: How does increasing the "Weight" (Intensity) affect the "Total Work" (Volume)? Does a higher weight lead to a drop in total volume, or is the progression linear?
- Optimization: Based on the data from March 13 to April 5, which exercise requires more focus to maintain a balanced strength?

C. Data Collection and Data Structure

1. Selected Personal Data Type

The selected data is Quantitative Physical Performance Data, specifically focusing on cumulative resistance training metrics. This is that tracks the total mechanical output per exercise.

2. The Prompts Used

Cumulative Volume: Instead of recording a single set, the data has the aggregate performance for each exercise. For example, if an exercise consists of 3 sets of 12 repetitions at 50kg, the total repetitions (36) and the weight (50kg) are used to calculate the session's impact.

Exercise Standardization: Only repetitions performed with a consistent weight and proper form are included. Warm up sets are not included.

Metric Precision: Weights are recorded in kilograms (kg).

3. Time Period Tracked

The data was consistently tracked over a 24-day period:

Start Date: March 13, 2026

End Date: April 5, 2026

Total Frequency: 11 sessions (3 sessions per week).

4. Data Collection Method

The data was collected using a Manual Entry Method. Performance was initially noted during the workout and later transferred into Microsoft Excel. The spreadsheet uses formulas to automatically calculate the "Total Work (Volume)" based on the reps and weights entered.

5. Dataset Sample

date	Hack Squat			Machine Chest Press			Cable Row			Daily Total Volume
	Weight in kgs	Repetitions	Total Work (Volume)	Weight in kgs	Repetitions	Total Work (Volume)	Weight in kgs	Repetitions	Total Work (Volume)	
13.Mar	70	24	1680	50	24	1200	35	36	1260	4140
15.Mar	70	27	1890	50	24	1200	35	45	1575	4665
18.Mar	75	24	1800	55	24	1320	40	36	1440	4560

6. Data Dictionary

The variables in the dataset are defined as follows:

Variable	Type	Description
Date	Date/Time	The calendar date of the training session.
Exercise	Categorical	The name of the specific exercise.
Weights (kg)	Numeric	The weight used for the sets performed (Intensity Metric).
Total Repetitions	Numeric	The sum of all repetitions performed across all sets for that specific exercise and weight.
Total Work (Volume)	Numeric	Calculated field (Weights times Total Repetitions) representing the total volume.

D. BI Workflow Diagram

Problem

- Phsical progression is often guesswork
- Need data-driven approach to measure "Progressive Overload" and optimize strength/volume

Data Collection

- Manual tracking during training sessions
- Transferring logs from mobile phone to a structured file (MS Excel)

Data Preparation

- Structuring raw data in Excel
- 3 compound exercises are recorded per session

Metrics

- Intensity, weights in kg
- Total repetitions per exercise
- Total Work (Volume) per session and exercise

Visualization

- Tool: MS Excel
- Chart Types: Line Chart, Bar Chart, Scatter Plot, Box Plot

Insights

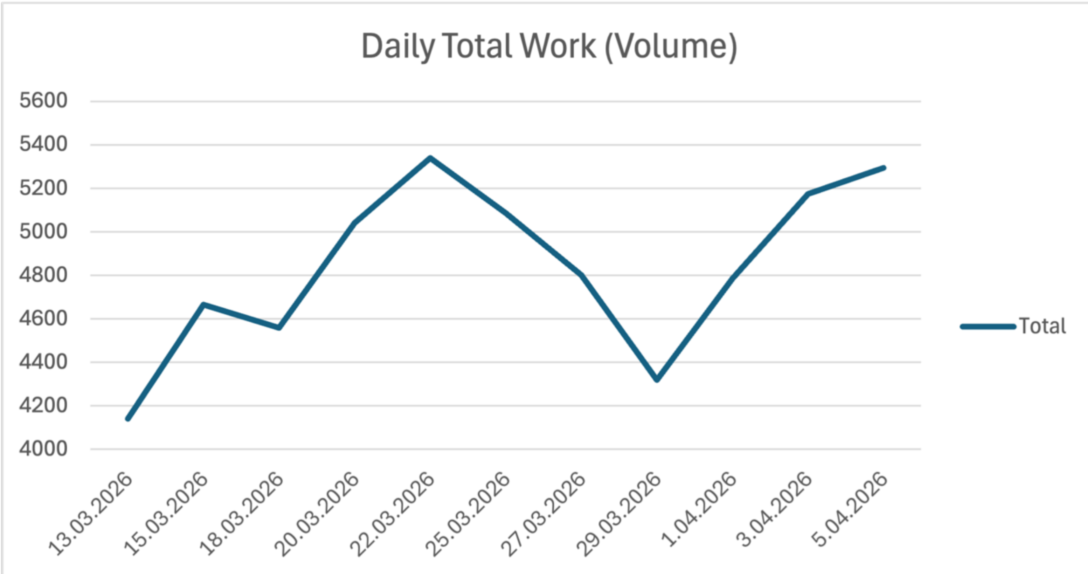
- Identifying strength imbalances
- Detecting peak performance sessions

Recommendation

- Targeted weight increases for specific exercises. Planning "rest" based on total volume stress.

E. Visualization Portfolio &

F. Annotation and Explanation for Each Visualization



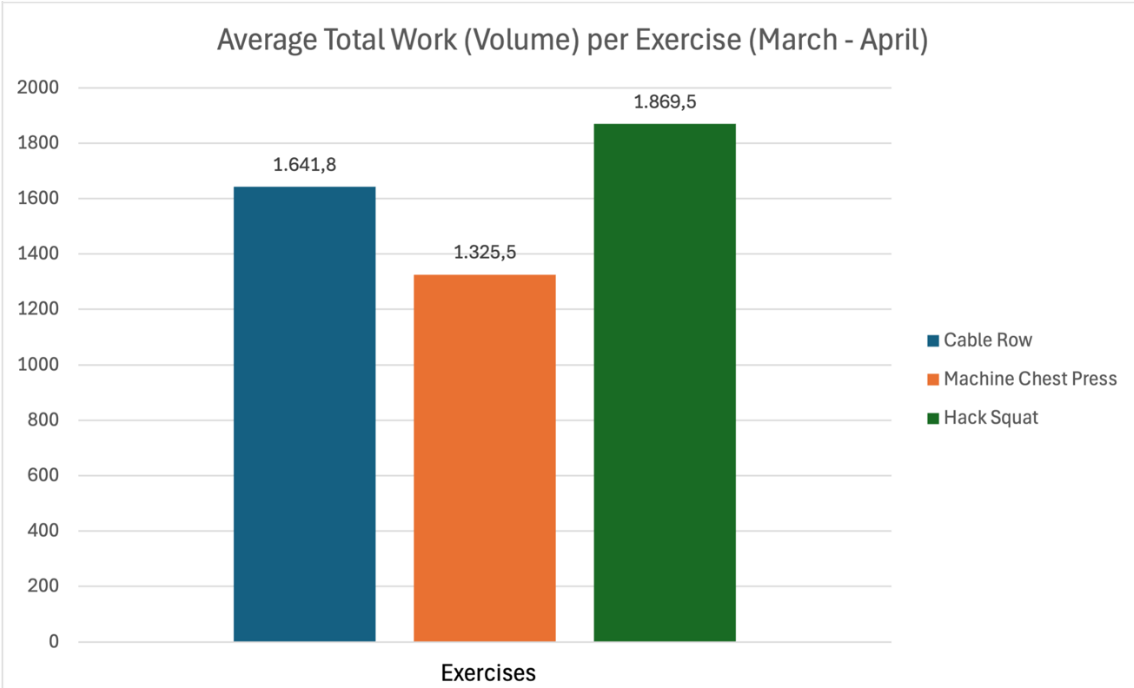
Visualization 1: Daily Total Work Volume Trend Analysis

Research Question: What is the compound growth rate of "Daily Total Work(Volume)" over the 24-day period?

Why this chart type is appropriate: Time-Series Line Chart is specifically chosen here to visualize the velocity of growth and identify specific dates where "Total Work (Volume)" deviated from the trend.

Interpretation & Insights: The chart shows a clear 28% increase in total work between March 13 and April 5, confirming a steady upward trend in performance. After an initial climb that reached a peak of 5,300 kg, there was a noticeable drop on March 29, but performance levels quickly recovered within just one week.

Key Takeaway: The overall trend indicates steady and healthy progress. However, the performance dip in late March highlights that recovery is a vital part of the process. To maintain high performance levels and prevent future declines, prioritizing recovery is just as essential as the training itself.



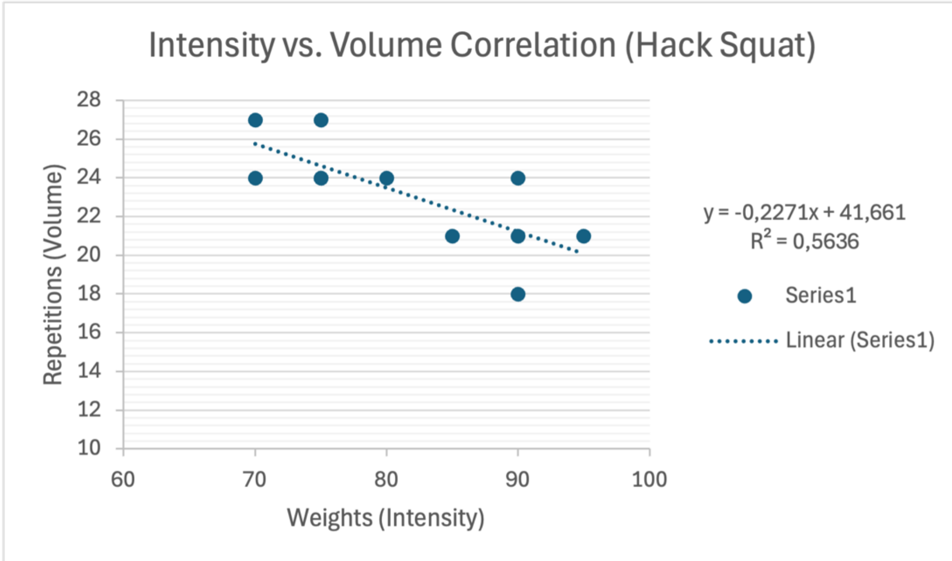
Visualization 2: Average Operational Capacity per Movement

Research Question: Which exercise yields the highest average ‘Total Work(Volume)’, and how do the lower-body vs. upper-body units compare in baseline performance?

Why this chart type is appropriate: Clustered Column Chart allows for a direct side-by-side comparison of distinct movements to identify dominant drivers of ‘Total Work volume’.

Interpretation & Insights: The comparison chart reveals that the Hack Squat operation maintains the highest average volume (1,800+ kg), outperforming the upper-body movements by a significant margin. This confirms that the lower-body exercise has a much higher baseline capacity for mechanical load compared to the chest and back exercises.

Key Takeaway: The data identifies the Hack Squat as the primary contributor to total work volume. To improve overall performance, focusing on leg exercises is expected to yield the most significant results. While upper-body movements show lower volume, they remain essential for maintaining muscular balance and structural integrity.



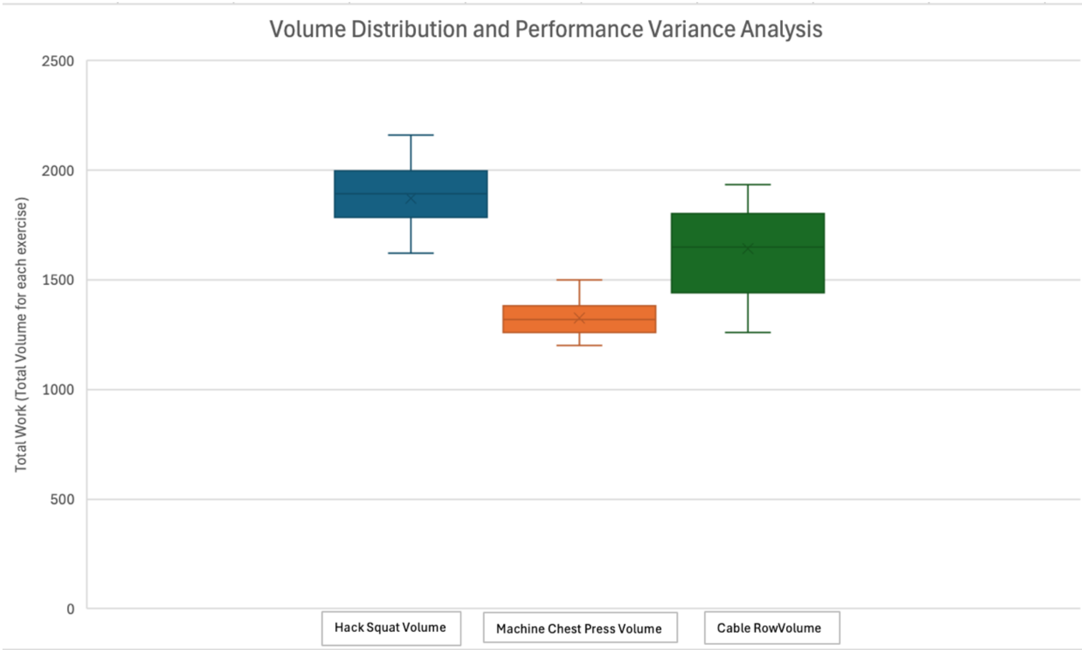
Visualization 3: Correlation Between Intensity and Volume

Research Question Addressed: How does an increase in "Difficulty" (Weight) impact the "Volume" (Repetitions), and how predictable is this relationship within the Hack Squat exercise?

Why this chart type is appropriate: Scatter Plot allows us to visualize the relationship between two continuous numeric variables and calculate the Coefficient of Determination (R^2), which quantifies how much of the variance in repetitions is explained by the change in weight.

Interpretation & Insights: The scatter plot demonstrates a moderate negative correlation (R^2 = 0.56) between weight and repetitions. As weight increases, the number of repetitions naturally decreases, which is an expected outcome. However, the steady slope proves that the physical system is handling the extra load effectively without a major performance crash. While weight is the primary reason for these changes, approximately 44% of the results are still influenced by external factors such as sleep quality or training time.

Key Takeaway: The system is achieving Progressive Overload with high efficiency. The fact that the volume does not "collapse" as weights increase proves a high level of Mechanical Adaptation.



Visualization 4: Volume Distribution and Performance Variance Analysis

Research Question Addressed: How consistent are the different exercises, and which exercises show the most change?

Why this chart type is appropriate: A Box and Whisker Plot is a good choice because it shows how the data is spread out. Unlike a bar chart that only gives an average, it shows the middle value, how the data is grouped, and the full range. This helps us see how stable the performance is and spot any unusual values or changes.

Interpretation & Insights: This analysis reveals different performance patterns across the three exercises. While the Hack Squat shows the highest overall work volume, it also has a wider range of results, indicating that performance is more sensitive to daily energy levels or fatigue. In contrast, the Machine Chest Press is the most reliable exercise; its steady and predictable results show very little change between sessions. Finally, the Cable Row exhibits significant fluctuations, suggesting that this movement needs more consistency to stabilize performance levels over time.

Key Takeaway: The results suggest that both the Cable Row and Hack Squat need more focus on consistency. In performance analysis, making results steady is just as important as increasing the total amount of work. Maintaining this consistency makes it easier to plan future progress and helps prevent the risk of overtraining.

G. Insights and Recommendations

1. Summary of Discoveries:

The 24-day tracking period shows that the body is in a phase of rapid growth. The most important finding is a 28% overall increase in total work volume. While all exercises improved, the data shows a clear difference between high-volume exercises like the Hack Squat and highly consistent ones like the Machine Chest Press. Additionally, the quick recovery after the performance drop in late March proves that the system is resilient, which is a key factor for maintaining progress over the long term.

2. The Most Important Pattern: Progressive Overload Efficiency

The most important trend identified is the balance between weight and repetitions. Typically, lifting heavier weights can lead to a significant drop in total work done. However, the data shows that while repetitions decreased slightly as the weights rose, the total work continued to trend upward. This pattern confirms that the body is adapting effectively, allowing the performance to improve without a loss in overall results.

3. Evidence-Based Recommendations:

Improving Consistency: The fluctuations seen in the Cable Row and Hack Squat suggest a need for more consistent rest periods and exercise speed. Making these sessions more uniform will lead to more predictable progress.

Strategic Growth: Since the Hack Squat is the main contributor to the total volume, it should remain the primary focus.

Managing Fatigue: The performance drop on March 29th shows the importance of planned recovery. Adding a lighter training week or increasing rest when the total workload becomes very high will help prevent burnout and keep performance steady over time.

4. Next Steps for a Professional Application

If this was a professional project for a pro athlete, a coach or manager would take the following steps:

Collect More Data: To better understand the 44% of results that are not explained by weight, more factors would be tracked. This includes collecting data on sleep quality, daily calories, and recovery levels to get a fuller picture of performance.

Future Forecasting: The progress from these 11 sessions would be used to predict performance for the next six months. This helps in planning ahead for important events or identifying when new exercise might be needed.

H. Reflection and Limitations

1. Data Quality and Inconsistencies

The data quality is generally high due to the structured nature of weight training; however, while "Total Work" was calculated mathematically, the quality of execution (form) was not measured. A session with higher volume but poor form could technically show "growth" on a chart while actually representing a decrease in exercise efficiency.

2. Sample Size Limitations

With only 11 sessions recorded over 24 days, the amount of data is relatively small. In a professional setting, this limited information makes it difficult to make highly accurate long-term predictions.

3. Possible Bias in Self-Tracking

The tracking of this data for a university project likely influenced the final results. Knowing that these training sessions would be turned into charts may have encouraged the athlete to push harder than they usually would during a normal workout. As a result, the growth shown in the data might be slightly higher than what would happen in a regular, untracked training routine.

4. Privacy and Ethical Considerations

This project involves personal health information, which is sensitive. In a professional business setting, such as a fitness app company, this data would be protected by strict privacy laws.

5. Future Improvements

If more time and resources were available, the following steps would be taken to make the analysis more detailed:

Tracking More Factors: Since some performance changes are still unexplained, adding information like sleep, diet, and stress levels would help clarify why results fluctuate.

Comparing Results: Adding data from other people would help create a standard to see how a typical person's progress compares to others.